JAEHYUN JEON

+82 10 8590 0664 | eric0664@kaist.ac.kr | jaehyunj@stanford.edu | Google Scholar

EDUCTATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2021 – Present
- B.S. Physics (Double Major : Electrical Engineering) see courses

Stanford University Jun. 2026 – Present
- Visiting Research, supervised by prof. Jonathan Fan in Electrical Engineering

Harvard Medical School Jun. 2025 – Aug. 2025
- Visiting Research, supervised by prof. Brett Bouma in Wellman Center

Daegu Science High School Mar. 2018 – Feb. 2021
- Specialized high school for gifted students in mathematics and science
- Received highest honor in national physics olympiad - general physics track.

PUBLICATIONS

5. **J. Jeon**†, H. Kim†, M. Jang, S. Menabde**, M. S. Jang*, "Selective Excitation of Higher-Order Hyperbolic Phonon Polaritonic Mode", ***in preparation***
4. C. Park, **J. Jeon**, M. Jang* "Active Mid-Ifrared Rotating 3D Beam Switching via Freeform Germanium Metasurfaces", ***in preparation***
3. **J. Jeon**†, T. Dai†, J. Fan*, "Connected Manifolds of Bound States in the Continuum for Continuous Superchirality Control", ***in preparation***
2. **J. Jeon**†, C. Park†, D. Heo, H. Chung, M. S. Jang*, "Inverse design of nanophotonic color router robust to oblique incidence", ***Advanced Optical Materials*** 14.4 (2026): e01697 [link] [video]
1. C. Park†, S. Lee†, T. Lee, J. Kang, **J. Jeon**, C. Park, S. Kim, H. Chung*, M. S. Jang*, "Towards subwavelength pixels: nanophotonic color routers for ultra-compact high-efficiency CMOS image sensors", ***Journal of Optics***, 26(9), 093002, (2024) [link]

HONORS and AWARDS

Undergraduate Visiting Research Program, Stanford University Summer, 2026
Outstanding Individual Research Student 2024 – 2025
MIT-Harvard Summer Institute in Biomedical Optics Summer, 2025
KAIST International Research Participation Fellowship Apr., 2025
Excellence in Poster Presentation Award Jan. 2025
KAIST-MIT Quantum Winter School Jan. 2025
Academic Excellence Scholarship Fall, 2024
- Merit-based scholarship for second-highest ranking in the department of physics

Dean's List Fall, 2024
- Demonstrated the best performance in academics, especially in major coursework

RESEARCH EXPERIENCES

Electromagnetics & Sustainability Group, Stanford Jun. 2026 - Present

Advised by *Prof. Jonathan A. Fan*

- **Bound States in the Continuum in High-Dimensional Shape Space**
  - Theoretically derived conditions on arbitrary shape-perturbation paths that yield $Q \sim |\delta|^{-2\mu}$ scaling for general geometric parameterizations, and established exact BICs as locally connected constraint manifolds in high-dimensional shape space.
  - Developing connected exact-BIC manifolds for continuous eigenmode engineering and controlled high-Q vectorial quasi-BIC generation, with near-field superchirality as a target functionality.
  - Parameterized level-set fields using geometric modes obtained through singular value decomposition, and combining adjoint-gradient to navigate the BIC manifold.

Quantum Nano Optics Laboratory, KAIST Dec. 2022 – Present

Advised by *Prof. Min Seok Jang*

- **Selective Excitation of Higher-Order Hyperbolic Phonon-Polaritonic Modes**
  - Developed an analytic framework based on the transfer matrix method to describe modal behavior, capturing mode conversion among image-polariton, phonon-polariton, and radiation modes.
  - Inverse-designed an Au grating beneath hBN to selectively excite the higher-order hyperbolic phonon-polariton mode. Validating the design through SNOM-based experiments.
- **Analytical and Inverse Design Framework for Moiré Fourier Photonic Surfaces**
  - Developed an analytical Floquet–Fourier framework for computing band structures of conductivity-modulated Fourier surfaces and commensurate moiré superlattices.
  - Designed a Fourier-harmonic inverse-design strategy for global bandgap opening by mapping band-edge eigenstates to reciprocal-space scattering channels and optimizing symmetry-adapted harmonics for cavity formation.
- **Fundamental Limit of Tunable Nanophotonic Devices and Time Photonic Crystals**
  - Applied a Maxwell-constrained QCQP framework to establish fundamental performance bounds for thermo-optically tunable metasurfaces with 3D diffraction-order control to compare with inverse designed devices.
  - Extended the fundamental-limit framework from binary spatial metasurfaces to binary photonic time crystals, and derived bounds on the achievable momentum-bandgap width and Floquet amplification rate under bounded temporal modulation.
- **Inverse Design of Freeform Metasurfaces in High Degree of Freedom**
  - Designed nanophotonic color router via gradient-based inverse design, achieving high performance under oblique incidence where prior normal-incidence designs suffered critical degradation.
  - Expertise in large-scale inverse design, specializing in gradient descent optimization using adjoint variable method and automatic differentiation.
- **Development of Fourier Neural Operator Based Surrogate Solver**
  - Implemented deep neural network as a surrogate solver predicting electromagnetic field distribution of freeform metamaterial across a broad wavelength range, based on Fourier Neural Operator. Utilizing the model, also implemented the inverse design framework for diverse functionalities.

Biomedical Optics Laboratory, Harvard Jun. 2025 - Aug. 2025

Advised by *Prof. Brett Bouma*

- **Shape sensing of multimode optical fiber via neural network** [link]
  - Predicting fiber shape from sparse transmission matrix data using an CNN-based neural network
  - Achieved over 98% prediction accuracy for 1m multimode fiber and analyzed the impact of fiber length and degrees of freedom.

INNERVERZ, Startup Company Summer, 2023

Advised by *Dr. Wonjong Ryu*

- **Vision AI Solutions for Creative Contents**
  - Based on CNN, developed neural network that recognizes video editing points and subtitles.
  - Developed a neural network for the analysis of facial expressions, angle variations, and human parsing.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C, Verilog HDL
Softwares and Tools: RCWA (Meent, Reticolo, S4), FDTD (Meep, Tidy3D, Lumerical), COMSOL
Language: English(Proficient, TOEFL : 105), Korean(Native)

PROJECTS

Design of Integrated Quantum Photonic and Mechanical Circuits

- Implemented quantum teleportation, CHSH inequality test, and MBQC using Qiskit and custom circuits.
- Designed and simulated photonic components: waveguides, MZIs, lattice filters, and ring resonators.

- Characterized MEMS cantilevers and surface acoustic wave (SAW) devices for micro-scale actuation, dynamic response analysis, and signal processing applications.

Nonlinear Optics & Solid State Physics Experiment
- Designed and analyzed an optical system for Second Harmonic Generation and Spontaneous Parametric Down-Conversion phenomena.
- Observed the Zeeman Effect using cadmium light source and analyzed spectral line splitting.
- Synthesized YBCO superconducting samples and measured their electrical resistivity across temperature variations.

Quantum Transport Simulation in Lattice Systems
- Reproduced quantum transport phenomena based on extended Hubbard model.
- Implemented time evolution of a 2D tight-binding lattice in the occupation-number basis, integrating Peierls-phase hopping and interaction terms.
- Validated gauge invariance and reproduced Bloch oscillations, Aharonov–Bohm interference, and synthetic Hall transport under multiple gauge configurations.

RELEVANT COURSEWORKS

Quantum Mechanics I & II, Solid State Physics I & II, Optics, Laser Optics, Nano Optics, Quantum Electronic Devices I: Interferometers and Fractional Particles, Electromagnetics I & II, Classical Electromagnetism I, Electronics Design Lab.<Design of integrated quantum photonic and mechanical circuits>, Physics Lab. III <Nonlinear Optics & Solid State Physics>, Introduction to Electronics Design Lab., Semiconductor IC Technology, Special Topics in Electronic Engineering I & II, Thermal Physics

**Including courses in progress*

EXTRACURRICULAR ACTIVITIES

Modern Rock Band
- Served as a band leader, vocalist, and guitarist, performing 28 times including several original compositions.

Volunteer Club
- Provided educational support to low-income student